

Hand Controller Rewrite

Testing Format Draft

Prepared as a discussion draft for the group before actual execution testing. This document contains only the proposed test data set, functional test cases, and nonfunctional test cases based on the current codebase analysis and the professor's instruction format.

Test Data Set

ID	Test Data	Category
TD-01	Normal launch environment: Windows, .venv ready, camera available, tuning.testing.json loaded	Valid
TD-02	Camera disabled or selected camera unavailable	Invalid
TD-03	Palm-facing open hand, one active hand visible	Valid
TD-04	Side-view or visually compressed hand	Invalid
TD-05	Thumb-index quick pinch and release	Valid
TD-06	Thumb-middle pinch	Valid
TD-07	Thumb-index hold for drag around 0.79s, 0.80s, 0.81s	Boundary
TD-08	ML toggle pose held around 0.44s, 0.45s, 0.46s	Boundary
TD-09	Thumb-ring hold around 0.34s, 0.35s, 0.36s	Boundary
TD-10	Two left tap cycles with interval around 1.49s, 1.50s, 1.51s	Boundary
TD-11	ML hold, undo, redo poses	Valid
TD-12	Finger hover over keyboard key then thumb-index pinch	Valid
TD-13	Two cameras connected; switch source while app is running	Valid
TD-14	Long continuous run for 15 to 30 minutes	Nonfunctional
TD-15	Launch timing observation: app window, minimize behavior, overlay appearance	Nonfunctional

Functional Test Cases

ID	Requirement	Steps	Test Data	Expected Result
TC-F-01	App launches successfully	Run .\run-tester.ps1; wait for control panel and overlay/live worker startup	TD-01	App opens without crash; control panel appears; live path starts

ID	Requirement	Steps	Test Data	Expected Result
TC-F-02	Camera preflight blocks bad camera state	Launch with unavailable or disabled camera	TD-02	App does not proceed silently into a broken state; fallback or warning is shown
TC-F-03	Mouse movement works only when allowed	Show palm-facing active hand, move hand slowly and then normally; repeat with side-view hand	TD-03, TD-04	Cursor moves when palm-facing and control is on; cursor stops or remains frozen when gating fails
TC-F-04	Left click works on quick pinch-release	Hover cursor on clickable target; do quick thumb-index pinch and release	TD-05	Exactly one left click is triggered
TC-F-05	Right click works on thumb-middle pinch	Hover cursor on target; do thumb-middle pinch	TD-06	Context menu or right-click action appears once
TC-F-06	Double click timing is handled correctly	Perform two quick left tap cycles; repeat near timing boundary	TD-05, TD-10	Inside boundary, double click is recognized; outside boundary, behavior becomes separate clicks or no double click
TC-F-07	Drag starts only after hold threshold	Hold thumb-index pinch at 0.79s, 0.80s, 0.81s; move hand while holding; then release	TD-07	Before threshold, no drag; at or after threshold, drag starts; release ends drag
TC-F-08	ML control toggle requires sustained hold and cooldown	Show toggle pose for 0.44s, 0.45s, 0.46s; repeat again within and after cooldown	TD-08	Below threshold, no toggle; at or after threshold, control toggles once; repeated trigger inside cooldown is blocked
TC-F-09	ML hold freezes movement and clicking	While in mouse mode, activate hold; try moving and clicking	TD-11	Mouse movement and click actions stop while hold is active
TC-F-10	Undo and redo work only in mouse mode	In mouse mode trigger undo and redo; switch to keyboard mode and repeat	TD-11, TD-09	In mouse mode sends Ctrl+Z and Ctrl+Y; in keyboard mode no undo or redo action should fire
TC-F-11	Side-view safety blocks press-like gestures	Use side-view hand and try left click, right click, drag start, and keyboard toggle	TD-04	Press-like actions are blocked or greatly reduced
TC-F-12	Keyboard mode toggle works at hold threshold	Hold thumb-ring pinch for 0.34s, 0.35s, 0.36s	TD-09	Below threshold no mode change; at or after threshold app switches between mouse and keyboard mode
TC-F-13	Keyboard typing works on alpha page	Enter keyboard mode; hover over letter keys; pinch to type	TD-12	Hovered key is highlighted; correct letter is typed
TC-F-14	Keyboard special keys and page switching work	Test ABC or 123, Shift, Caps, Backspace, Space, Enter, ESC, TAB	TD-12	Correct page switch or key output occurs; shift and caps state is reflected in behavior
TC-F-15	Camera refresh and live camera switching work	Refresh sources; switch to another connected camera while running	TD-13	Source list updates; app uses new camera without full app failure

Nonfunctional Test Cases

ID	Requirement	Steps	Test Data	Expected Result
TC-NF-01	Startup responsiveness	Launch app and observe how quickly window minimizes and overlay appears	TD-15	Launch feels responsive; minimize happens quickly; overlay or selfie appears without long visible freeze
TC-NF-02	UI responsiveness during preflight	While launch preflight or camera refresh is running, observe UI responsiveness	TD-15	Main window does not hang; controls show pending state correctly
TC-NF-03	Runtime stability	Run app continuously for 15 to 30 minutes with normal gesture use	TD-14	No crash, no severe lag buildup, no unrecoverable camera or overlay failure

ID	Requirement	Steps	Test Data	Expected Result
TC-NF-04	Reliability under fallback conditions	Start with missing camera or changed camera availability; observe fallback behavior	TD-02, TD-13	System fails gracefully, shows warning or fallback, and does not silently misbehave

Note: This PDF is a draft testing format for group review. Actual Pass or Fail results, observed behavior, and final status should be filled in after manual execution testing.